Chethan Ramakrishna Reddy

Mechatronics Engineer

chethan.reddy@gmail.com
https://chethanrreddy.github.io
+1 906 275 9969

Education

- **Michigan Technological University** — Houghton, MI, USA
  PhD in Mechanical Engineering — *2016 - 2022*
- **National Institute of Technology Karnataka** — Surathkal, India
  MTech (MS equivalent) in Mechatronics Engineering — *2011 - 2013*
- **Visvesvaraya Technological University** — Belgaum, India
  BE (BS equivalent) in Mechanical Engineering — *2007 - 2011*

Educational Details

1. PhD at Michigan Technological University – Transcript
   - **CGPA – 3.77/4.00**
   - PhD Dissertation: **Model Predictive Control of Energy Systems for Heat and Power Applications** – Dissertation
     - Aim of the project was to design a model predictive control (MPC) framework for energy systems.
     - The energy systems considered were (i) building heating, ventilation & air-condition (HVAC) system integrated with a micro-scale concentrated solar power (MicroCSP) system; and dual fuel internal combustion engine (ICE) integrated with a waste heat recovery (WHR) system.
     - The project involved plant model development, validation, MPC framework development, and MPC framework debugging. Tools used were MATLAB/Simulink, and engineering equation solver (EES).
2. MTech at National Institute of Technology Karnataka – Transcript
   - **CGPA: 8.37 / 10, US equivalent CGPA: 4.00 / 4.00**
   - Laboratory overview (Apr 2013)
     - Three day laboratory (Automotive Electronics lab, sponsored by RBEI) overview to fellow graduate students
   - Mini Project: Rotary Encoder using an 8051 Micro-Controller
     - Aim of the project was to reduce the cost of available rotary encoders
     - An incremental rotary encoder was designed and built using a stepper motor
     - Every step by the stepper motor gives a pulse, which was amplified and fed into the micro controller

  - The micro controller was programmed to show the angle turned by the stepper motor by an LED/LCD display.
- MTech project/thesis: **Development of Automotive Thermo-Electric Generator (ATEG)** – Thesis
  - Aim of the project was to recover waste heat energy from the exhaust of an Internal Combustion (IC) engine driven automobile and convert a part of it to electricity (useful form of energy) by the thermoelectric or Seebeck effect.
  - Project involved a detailed simulation model built in MATLAB/Simulink environment and an overview on the control strategy to realize ATEG. Model validation was done by giving the model inputs from the engine test bench data (measurements).

3. BE at Visvesvaraya Technological University – Transcript

- **Grade – 73.18%, US equivalent CGPA: 3.73 / 4.00**
- Head of the Print media team for Mechanical Engineering Department (2009-2011)
  - Responsible for writing columns related to events in the department and send to print media.
- Final semester project: **Design and Fabrication of Boundary Layer Turbine as a Potential Automotive engine (Compressed Air as Fuel)** – Report
  - Based on invention by Nikola Tesla in early 20th century
  - Aim of the project was to develop a compressed air engine for an automobile.
  - The project involved design and analysis of a rotary boundary layer turbine as an automobile engine. Tools used were Solidegde.

Work Experience

1. **Engineer – Ford Motor Company** — SF Bay Area/Remote, USA
   Model Based Calibration Optimization for xEV — *Jan 2022 - Present*
2. **Research Assistant – Michigan Technological University** — Houghton, MI, USA
   Energy Mechatronics Laboratory — *May 2017 - Jan 2022*
3. **Teaching Assistant – Michigan Technological University** — Houghton, MI, USA
   Mechanical Engineering – Engineering Mechanics Department — *Aug 2019 - May 2020*
4. **Product Intern – Constructis** — MI/SC/Remote, USA
   System Engineering — *Aug 2020 Dec 2020*
5. **Automotive Controls Intern – Halla Mechatronics** — Bay City, MI, USA
   Motor Controls Group — *Jan 2019 - May 2019*
6. **Teaching Assistant – Michigan Technological University** — Houghton, MI, USA
   Mechanical Engineering – Engineering Mechanics Department — *Aug 2017 - Dec 2018*
7. **Senior Engineer – Robert Bosch India** — Bangalore, India
   System Engineering Group (Hybrid Systems & E-Mobility) — *Oct 2015 - Aug 2016*
8. **Engineer – Robert Bosch India** — Bangalore, India
   Modeling & System Simulation Group — *Aug 2013 - Sep 2013*

9. **Intern – Robert Bosch India** Bangalore, India
Modeling & System Simulation Group *Jun 2012 - Mar 2013*

10. **Teaching Assistant – National Institute of Technology Karnataka** Surathkal, India
Mechanical Engineering Department *Jan 2012 - May 2012*

Work Details

1. Engineer at Ford Motor Company

    - Model Verification - The aim of this project was to verify a developed electrified vehicle model (powertrain and energy management focused) when there was no test data to validate the model. The approach used was to develop pass/fail criterion to subject the model based on historical data and system knowledge. This work was done in Matlab/Simulink.
    - Model Based Calibration Optimization - The aim of this project is to utilize the electrified vehicle model and provide an optimal range of the calibration data-set and hence reduce the testing cost/time/effort. This work was done in Matlab/Simulink in high performance computer (HPC); which runs on Linux OS.
    - Model Speed Improvement - The aim of this project was to improve the compilation and run the run time of the developed electrified vehicle model (powertrain and energy management focused). The approach used was to optimize the control model, plant model in terms of step size, sample time, and solver. A prerequisite to this was to convert the model(s) exported from other tools to Matlab/Simulink; from s-function to co-simulation Functional Mockup Unit(s) (FMU).
    - Multi Auto Correlation Post Processing - The aim of this project was to provide quantifiable meaning to the multiple signal correlations (Model out vs. Plant measurements) of multiple calibrations. This work was done in Matlab/Simulink.

2. Research Assistant at Michigan Technological University

    - Research in the Energy Mechatronics Laboratory at Mechanical Engineering - Engineering Mechanics department.
    - Research on **Model Predictive Control of (i) Building HVAC System with Solar Energy Integration, and (ii) Internal Combustion Engine with Waste Heat Recovery.**

3. Teaching Assistant at Michigan Technological University

    - Lab instructor for two courses handling a total of about 45 students (3 Laboratory sessions) in Fall 2019 & Spring 2020 terms. The first course focuses on dynamics and control of mechanical systems, the second course focuses on introductory manufacturing processes. Both courses consist mostly of senior undergraduate students in Mechanical engineering.
    - Student reviews of 3.71/5 for Fall 2019.

4. Product Intern at Constructis

    - System (Mechanical, Electrical, and Software) Definition, Design, and Analysis for the Roadway Kinetic Energy Recovery Pilot Product.
    - Had to cut short due to budget delays and I had to travel to Ireland due to family emergency.

5. Automotive Controls Intern at Halla Mechatronics

    - Internship task was to model, validate and simulate a vehicle with electrically assisted power steering (EPS) system developed by Halla Mechatronics.
    - The vehicle dynamics was modeled in CarSim. The proprietary EPS system along with the EPS controller of Halla Mechatronics was modeled in Matlab and Simulink.
    - The objective was to co-simulate the plant model with the controller model in order to simulate test cases to understand the controller and vehicle behaviour.

6. Teaching Assistant at Michigan Technological University

    - Lab instructor for about 45 students (3 Laboratory sessions) in Fall 2017, Spring 2018, and Fall 2018 terms. The course focuses on dynamics and control of mechanical systems, and consists mostly of senior undergraduate students in Mechanical engineering.
    - Student reviews of 3.97/5, 4/5, and 3.98/5 for Fall 2017, Spring 2018 and Fall 2018, respectively.

7. Senior Engineer at Robert Bosch India – Service letter

    - Active Noise Cancellation and Enhancement – The aim was to develop a product for automobiles (two wheelers, passenger cars, off road vehicles etc.) capable of cancelling or enhancing sound at the exhaust of the vehicle. This is achieved actively. We cancel sound if the vehicle exhaust is too loud or enhance it to make a passenger car sound like a F1 car (for example). I was responsible fully on the technical implementation of the project (from vehicle noise measurement, mechanical setup of sensor and actuator to electronic control unit development). I was successfully able to develop two demonstrators on test vehicle (a two-wheeler). One with a pure static analog hardwired solution for active noise cancellation and an adaptive software solution for cancellation/enhancement both (a switch to opt either of the two). The software tools I used for this project are MATLAB/Simulink and DSPACE MicroAutoBox 2 (rapid prototyping electronic control unit (ECU)).
    - Bosch Boost Recuperation System (BRS) Simulation – BRS is an advance system for mild hybrid and CO2 reduction by Bosch, one step ahead of conventional start/stop systems. It provides start/stop, coasting, brake recovery and also boosting using a controller and an electrical machine (acts as a motor and also generator) coupled to the Engine. BRS is typically a 48 Volt system (i.e. consisting of a 48 Volt electrical machine) in developed markets like the Europe and America but for emerging markets such as India, a 12 Volt BRS (i.e. consisting of a 12 Volt electrical machine) is more appropriate due to cost and smaller Engines. So, my task was to develop both the control (mainly because control strategies for European and Indian market will be markedly different) and plant model to analyze the advantages of a 12 Volt BRS and 48 Volt BRS for Indian market and the different vehicle segments of the Indian market. One more important task in this project was to validate the pros and cons of using a Supercapacitor/Ultracapacitor/Hybrid-Supercapacitor as an alternate ESS (Energy Storage System) to the Li-Ion Battery for Indian Market. This work was done partly in GT-Suite and partly in MATLAB/Simulink.
    - Automobile Waste Heat Recovery using Thermoelectric Generators This is mainly a self-initiation project. In addition to the work I had done during my internship at Bosch, I have further built competency, interacted with thermoelectric generator manufacturers, studied use cases in the Automotive domain, and demonstrated a use case on a two-wheeler.

8. Engineer at Robert Bosch India

- HIL Plant Model Development – A Hardware in Loop setup (HIL) contains a control model (to be tested in its target hardware) and a plant model. My role was in developing, calibrating, and generate code of the plant model (in this case an Automobile) in MATLAB/Simulink.
- Model-Based Testing – The aim of this project was to develop control model in a simulation environment (MATLAB/Simulink) and then generate code out of it. Testing and validation (of the control logic) is done, by Model in Loop testing (MIL - integrating the control model with plant model and testing by feeding the model with standard test cases) and Software in Loop testing (SIL - generating code out of the control model and integrating it with the same plant model and feed the co-simulation model with standard test inputs), in the development phase itself thus saving time and resources. I was responsible for all the technical implementation for this pilot project.
- Model-Based Design & Calibration – There are two aims with this project, they are
  - Built highly accurate plant models and use it to pre-calibrate the Maps/Tables in an Electronic Control Unit (ECU)
  - Built highly accurate plant models and use it to make design decisions (for ex: how big a component is required? How best to connect it? etc.)

  The outcome of this project is lesser dependency on Test Bench, faster time to market; leading to cost savings. The project is carried out in GT-Suite and MATLAB/Simulink.
- Virtual Hardware – The aim of this project is to be able to model/simulate the whole embedded system in a PC environment. That includes an accurate microcontroller model (purchased for microcontroller manufacturers), ASIC models (usually the electrical drivers), control model (software inside the microcontroller) and plant model (to give various sensor and receive actuator signals from the control model). The use case of this pilot project will benefit the organization in time, money and resources. My role was to completely develop and validate the plant model. This included calibrating the model to the use case, develop missing models and validate the same with integrating with the rest of the system. MATLAB/Simulink was used to develop, calibrate and generate code out of the model.

9. Intern at Robert Bosch India – Certificate

   - Practical Training – Report
     - Built a Stepper Motor interface model (MATLAB/Simulink environment) as an Idle Air Control Actuator for a Fuel Injected Gasoline I.C. Engine
     - Modeled and simulated a complete Air Intake system (including cylinder breathing dynamics) considering a Gasoline 2-cylinder Engine – in MATLAB/Simulink environment.
   - MTech project/thesis – Please see education details

10. Teaching Assistant at National Institute of Technology Karnataka

    - Lab instructor for about 60 students (3 Laboratory sessions) in Fall 2012 term. The course focuses on computer aided engineering drawing of mechanical systems, and consists mostly of junior undergraduate students in Mechanical engineering.

Publications

- **C. R. Reddy**, V. Bonforchi Vinhaes, J. D. Naber, R. D. Robinett III, M. Shahbakhti, **'Exergy based model predictive control of an integrated dual fuel engine and a waste heat recovery system'**, in *Control Engineering Practice*, 2023. Link
- **C. R. Reddy**, V. Bonfochi Vinhaes, J. D. Naber, R. D. Robinett III, M. Shahbakhti, **'Model predictive control of a dual fuel engine integrated with waste heat recovery used for electric power in buildings'**, in *Optimal Control Applications and Methods*, 2022. Link
- **C. R. Reddy**, V. Bonfochi Vinhaes, R. D. Robinett III, J. D. Naber, M. Shahbakhti, **'Model Predictive Control of a Waste Heat Recovery System Integrated with a Dual Fuel Natural Gas-Diesel Engine'**, in *IEEE American Control Conference*, 2021. Link
- M. Toub, **C. R. Reddy**, R. D. Robinett III, M. Shahbakhti, **"Integration and Optimal Control of MicroCSP with Building HVAC Systems: Review and Future Directions"**, in *Energies*, Volume 14(3), 2021. Link
- **C. R. Reddy**, M. Shahbakhti, R. D. Robinett, and M. Razmara, **"Exergy-wise predictive control framework for optimal performance of MicroCSP systems for HVAC applications in buildings"**, in *Energy Conversion and Management*, Volume 210, 2020. Link
- M. Toub, **C. R. Reddy**, M. Razmara, M. Shahbakhti, R. D. Robinett III, G. Aniba, **"Model-based predictive control for optimal MicroCSP operation integrated with building HVAC systems"**, in *Energy Conversion and Management*, Volume 199, 2019. Link
- **C. R. Reddy**, M. Shahbakhti, M. Razmara, R. D. Robinett III, **"Optimal Exergy-wise Predictive Control for a Combined MicroCSP and HVAC System in a Building"**, in *IEEE American Control Conference*, 2019. Link
- **C. R. Reddy**, M. Toub, M. Razmara, M. Shahbakhti, R. D. Robinett, G. Aniba, **"Modeling and Optimal Control of micro-CSP and a Building HVAC System to Minimize Electricity Cost"**, in *ASME Dynamic Systems and Control Conference*, 2018. Link
- M. Toub, **C. R. Reddy**, M. Razmara, M. Shahbakhti, R. D. Robinett, G. Aniba, **"Model Predictive Control for MicroCSP Integration into a Building HVAC System"**, in *IEEE International Conference on Control and Automation*, 2018. Link
- **C. R. Reddy**, S. S. Rao, V. Desai, K. Ramachandran, **"Modeling of an Automotive ThermoElectric Generator (ATEG)"**, in *International Journal of Science and Research* (*IJSR*), 2013. Link

Technical Skills

- Model Based System Engineering.
- Modeling, Simulation, App Design, Data Analysis, and Code Generation in Matlab/Simulink.

- Modeling, Simulation, and Data Analysis in Python.
- System Know how/Modeling of thermal, mechanical, hydraulic, pneumatic, electronic and electric sub-systems. 0D, 1D, 3D Modeling Techniques.
- System Simulation (GT Suite, AVL, CarSim), Co-Simulation (CS) and Model Exchange (ME). FMI/FMU Standard.
- Numerical Optimization Techniques / CPU usage Optimization Techniques / Simulation Model Optimization Techniques. High Performance Computing, Parallel Processing.
- Model Based Optimal Control Design, Model Based Calibration Optimization, Model Based Predictive Control Design. Linear & Non-Linear Model Based Optimization.
- Optimization Solvers & Techniques. Linear & Non-Linear Control Theory. Machine learning & Genetic Algorithms.
- Model, Software, and Hardware in Loop (MiL, SiL and HiL) Model Development, Integration, Validation & Verification.
- Mechanical CAD.
- ECU prototyping/HiL testing (Motohawk, dSPACE).
- Scripting/Software languages - Matlab (Read & Write), Python (Read & Write), C/C++ (Read).

Certification Courses

- Completed "Evaluating Writing Training Program" as a Graduate Teaching Assistant in the Department of Mechanical Engineering – Engineering Mechanics, Michigan Technological University, Houghton, Michigan, USA. Certificate
  Highlights:
  - Was a five week course focussing on writing evaluations of students as a GTA
  - Each week, we focussed on a topic (like the importance of communication in engineering, having a "GRADING RUBRIC" for faster and fair evaluation, student accommodations, fairness to student of diverse backgrounds etc) concerning evaluating students writing
- Completed a practical and hands on course in Automobile Servicing and Maintenance in G.D. Naidu Charities, Coimbatore, Tamil Nadu, India. Certificate
  Highlights:
  - Was a three week course on Automobile servicing and Maintenance
  - We were given all the typical Automobile parts (Engine, Clutch, Gear Box, Differential, Suspension, Steering, etc.).
  - These were stripped to most basic level and we learnt to service (clean) and re-assemble them
  - Was a course designed to understand the parts of a typical Automobile, realize their physical implementation and appreciate their performance

- Completed a familiarization course in H.A.L. Aircraft division, Bangalore, Karnataka, India. Certificate
  Highlights:
  - Was a month long Training/Familiarization course in the H.A.L. Aircraft division plant
  - Each day we were assigned a department/shop to get to know and familiarize ourselves with what they are doing, which component of which Aircraft they are producing, number of employees etc
  - In the final week, we were assigned to Aircraft service bay and final assembly plants, where I got a chance to sit inside and explore Sukhoi, LCA (Light Combat Aircraft), Surya Kiran, Hawk (the then training and combat Aircrafts used by the Indian Air Force)

Achievements

- IEEE Senior Member from 2023
- Doctoral Finishing Fellowship for Fall 2021
- Outstanding Graduate Teaching Assistant Award based on Fall 2017 student reviews
- Demonstrated ANCE (Active Noise Cancellation and Enhancement) project on a two-wheeler and was appreciated by many top-level management/technical leads of our organization including the president of our organization (Robert Bosch India). Our project ANCE was covered by various media organizations (Link 1, Link 2, Link 3)
- Took an active part in College fests at both Pre-University and Undergraduate level
- Was the school captain in Class X

Review Panels

- Project Continuation Review Panel for Senior Design Projects at Mechanical Engineering - Engineering Mechanics Department at Michigan Technological University
  - "DCCV Housing Process Improvement" in Fall 2017
  - "Aftertreatment Body Joint Design" in Spring 2018
  - "Exhaust Aftertreatment Substrate Heater Design" in Spring 2018
  - "Sand Point Tower and Boardwalk" in Fall 2019

Languages

- English – Business fluent English (Read, write & speak).
- Indian Languages known – Telugu (mother tongue), Kannada, & Hindi.
- German (Basic Conversation skills) – 1A qualified.

Personal Details

- Date of Birth: 14 December 1989
- Sex: Male
- Marital Status: Married